A Test of exjobb_eit.typ

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Frontmatter

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AC Actually Confusing

CODE Computers Only Do Exactly (what you tell them)

EIT Engineering Is (never) Trivial

EMC Engineers' Major Complaint

IC Infinite Confusion

LED Let's Emit Dazzle

MOSFET Mostly Our Small Friend Enabling Transistor

PING Packet Is Now Going

"To mature as an electrical engineer is to realize everything is about impedances."

— Markus Törmänen, spring 2025

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Chapter 1

Introduction

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguique possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et collaudata est, cum id, quod maxime placeat, facere possimus, omnis voluptas assumenda est, omnis dolor repellendus. Temporibus autem quibusdam et.

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$$\operatorname{div} \boldsymbol{A} \coloneqq \lim_{\Delta v \to 0} \frac{\oiint_S \boldsymbol{A} \cdot d\boldsymbol{s}}{\Delta v} \tag{1.1}$$

$$\nabla \times \boldsymbol{B} = \mu_0 \left(\boldsymbol{J} + \varepsilon_0 \frac{\partial \boldsymbol{E}}{\partial t} \right) \tag{1.2}$$

The equations above are Equation (1.1) and Equation (1.2). This is Chapter 1 but below is Section 1.1 and below that is Section 1.1.1.

1.1 Test section

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut.

1.1.1 A section which won't appear in the outline

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut.

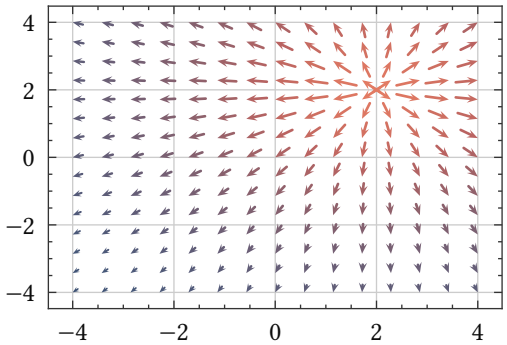

Figure 1.1: An example figure using the package Lilaq. [1]

1.2 Another test section

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguique possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et collaudata est, cum id, quod maxime placeat, facere possimus, omnis voluptas assumenda est, omnis dolor repellendus. Temporibus autem quibusdam et.[1]

Figure 1.2: A figure with a longer caption. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut.

[1] https://www.lth.se

Chapter 2

New Chapter with a Long Title that Spanns over More Than One Line

Figure 2.1: A figure together with another figure.

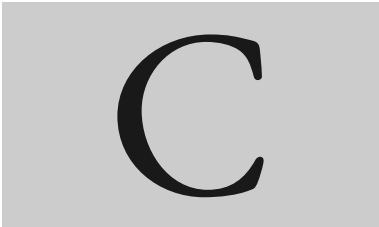

Figure 2.2: This figure can have a long caption and still fit if done this way. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do.

Above Figure 2.1 is next to Figure 2.2. By using the `#flexCaption(long caption, short caption)` function you can make flexible captions. You can also make longer/shorter headings! See Chapter 2.

2.1 New section

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguique possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et collaudata est, cum id, quod maxime placeat, facere possimus, omnis voluptas assumenda est, omnis dolor repellendus. Temporibus autem quibusdam et.

Table 2.1: A simple table.

t_0	t_1	t_2
y_0	y_1	y_2

2.2 Styling parameters

Table 2.2: Document types, their size and color.

	Headings	Sub-headings	Sub-sub-headings	Body	Secondary text	Tertiary
Size	`16pt` (Body), `42.52pt` (Chapter numbering)	`14pt`	`12pt`	`10pt`	`9pt`	-
Color						

Table 2.3: Fonts used.

	Body	Secondary text	Chapter numbering	Math	Raw
Font	EB Garamond	Lato	Libertinus Math	Libertinus Math	`Source Code Pro`
Usage	Main text	Headings, captions, header	Chapter and page number	Equations, inline math	Code blocks

Chapter 3

A word on Typst referencing

In References you can find links. [2] provides a guide for past LaTeX users on the differences between the two typesetting languages, including what's native or not native, which macros map to which function etc. [3] and [4] links to information on how to cite and reference different sources. Typst includes support for both its native YAML-based format and BibTeX. [5] gives a link to the Typst Universe. On this page you can find packages which aid in your workflow, both in terms of styling and automation. For example Figure 1.1 was made using lilaq, a package used to make graphs inside the Typst environment. Another good package is Zap which is used to make circuit diagrams much like TikZ. [6]

For documentation on all functions, markup- and styling commands, see [7].

References

[1] "Lilaq - Typst Universe." [Online]. Available: https://typst.app/universe/package/lilaq/

[2] "Guide for LaTeX users." [Online]. Available: https://typst.app/docs/guides/guide-for-latex-users/

[3] "Cite function." [Online]. Available: https://typst.app/docs/reference/model/cite/

[4] "Bibliography function." [Online]. Available: https://typst.app/docs/reference/model/bibliography/

[5] "Typst Universe." [Online]. Available: https://typst.app/universe/

[6] "Zap - Typst Universe." [Online]. Available: https://typst.app/universe/package/zap

[7] "Typst Documentation." [Online]. Available: https://typst.app/docs

Appendix A

Backmatter

```c
void insertionSort(int a[], int n)
{
    int i, key, j;
    for (i = 1; i < n; i++) {
        key = a[i];
        j = i - 1;

        while (j >= 0 && a[j] > key) {
            a[j + 1] = a[j];
            j = j - 1;
        }
        a[j + 1] = key;
    }
}
```

Listing A.1: Code snippet for insertionsort written in C.

Figure A.1: This is the last figure.